

Excel & Power BI for Data Analysis

A four-stage capability portfolio: ETL, data modeling, DAX & interactive dashboards

Macquarie University & Coursera datasets

Dashboard portfolio: [Excel and Power BI for Data Analysis](#)

Portfolio Summary

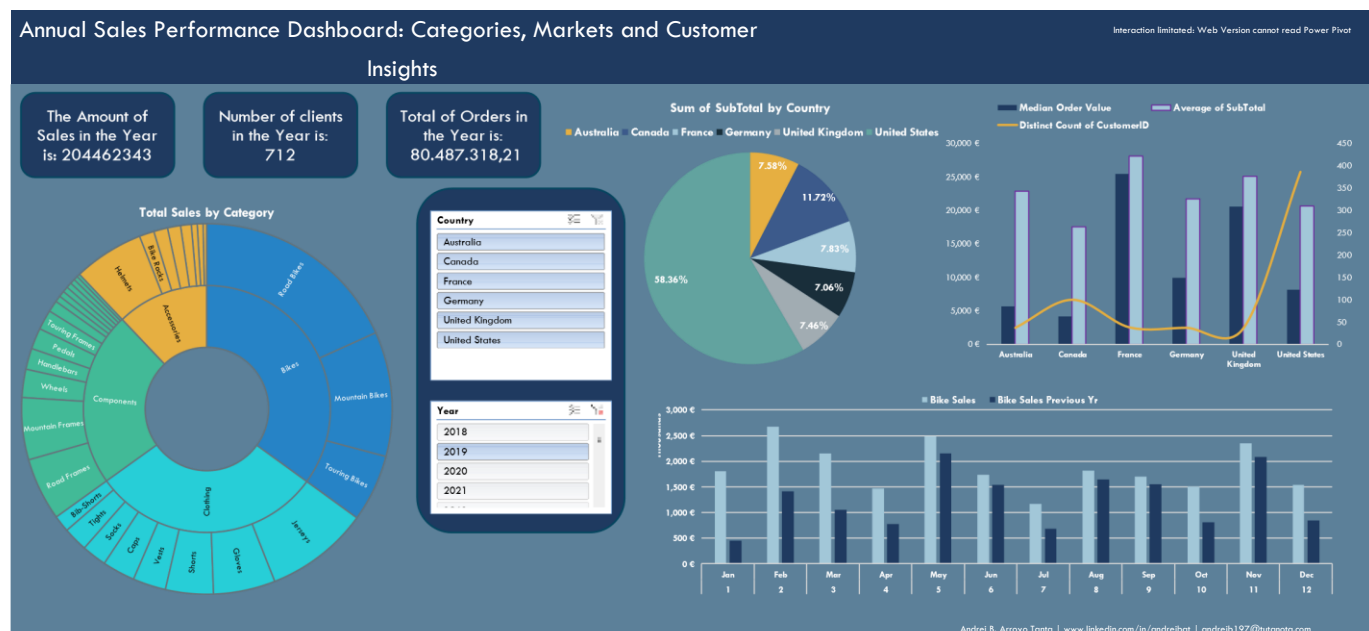
This portfolio documents my progression in **Excel** and its extensions through four projects built on datasets from the *Excel Skills for Data Analytics and Visualization* course at Macquarie University. Each stage adds technical depth — from **Power Query ETL automation** to **relational data modeling**, **DAX measures**, and **OLAP-style visualizations** — culminating in an enterprise-grade **Power BI** dashboard with time intelligence and dynamic fiscal calendars.

Note: for confidentiality, this document includes screenshots and a portfolio link only. Full file access is available on request.

PROJECT 1

Enterprise Sales Intelligence: Multi-Dimensional Data Architecture and Predictive Analytics with DAX

Designed a BI ecosystem based on relational models and **DAX functions**, transitioning from flat pivot tables to high-impact OLAP visualizations for year-over-year growth analysis, after optimizing data loading with Power Query.



Annual Sales Performance Dashboard — Categories, markets, and customer insights (\$204.5M sales, 712 clients, \$80.5M orders)

1. Business Need

In corporate environments with large data volumes, traditional analytics tools fail to correlate multiple sources (sales, goals, products, staff). The critical issue is metric fragmentation: the difficulty of comparing current vs. previous performance dynamically and hierarchically. Without a robust data model, companies cannot drill down without losing data integrity.

2. Strategic Importance & Impact

- ▶ **Time intelligence:** DAX functions like SAMEPERIODLASTYEAR let management evaluate organic year-over-year growth and distinguish real trends from seasonality.
- ▶ **Hierarchical information navigation:** hierarchies let users drill from macro category to specific product in a single click, speeding response to sales deviations.
- ▶ **Advanced visualization & OLAP flexibility:** converting pivot tables into CUBE functions breaks layout rigidity, enabling Sunburst charts linked to Slicers for richer UX.

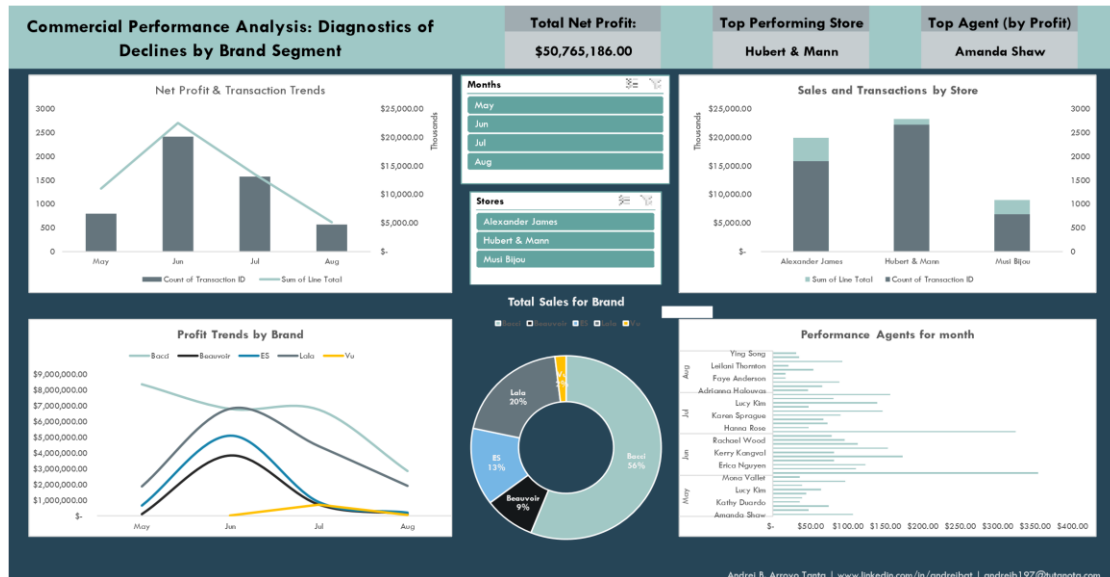
3. Technology Stack & Technical Competencies

- ▶ **Relational data modeling (Power Pivot):** 1:N relationships between fact and dimension tables, building a star schema for referential integrity and performance.
- ▶ **Metrics engineering with DAX:** complex CALCULATE measures to modify filter context, plus time-intelligence functions for comparative analysis.
- ▶ **OLAP architecture & cube formulas:** converted the dynamic model into member/cube value functions, enabling unconventional fully-interactive visualizations.
- ▶ **Metadata enrichment (hierarchies):** technical data categorization to surface insights through integrated hierarchical structures.
- ▶ **Query optimization (Power Query):** efficient file loading and pre-processing ensuring a clean, correctly-typed Power Pivot model.

PROJECT 2

Automated Retail Intelligence Hub: Scalability and Performance Analysis in a Business Chain

Designed an automated data ecosystem using **Power Query** that integrates multiple transactional sources, eliminating information silos and enabling real-time performance analysis for strategic decision-making.



1. Business Need

In the cosmetics distribution industry, agility is key. The core problem was data disconnection: sales arrived weekly as flat text files, while product, agent, and branch lists lived in separate, static, cluttered spreadsheets. This fragmentation created an error-prone manual process, wasting days of consolidation before brand performance or agent productivity could even be analyzed. The business needed a "single source of truth" able to absorb new weekly data with a single click.

2. Strategic Importance & Impact

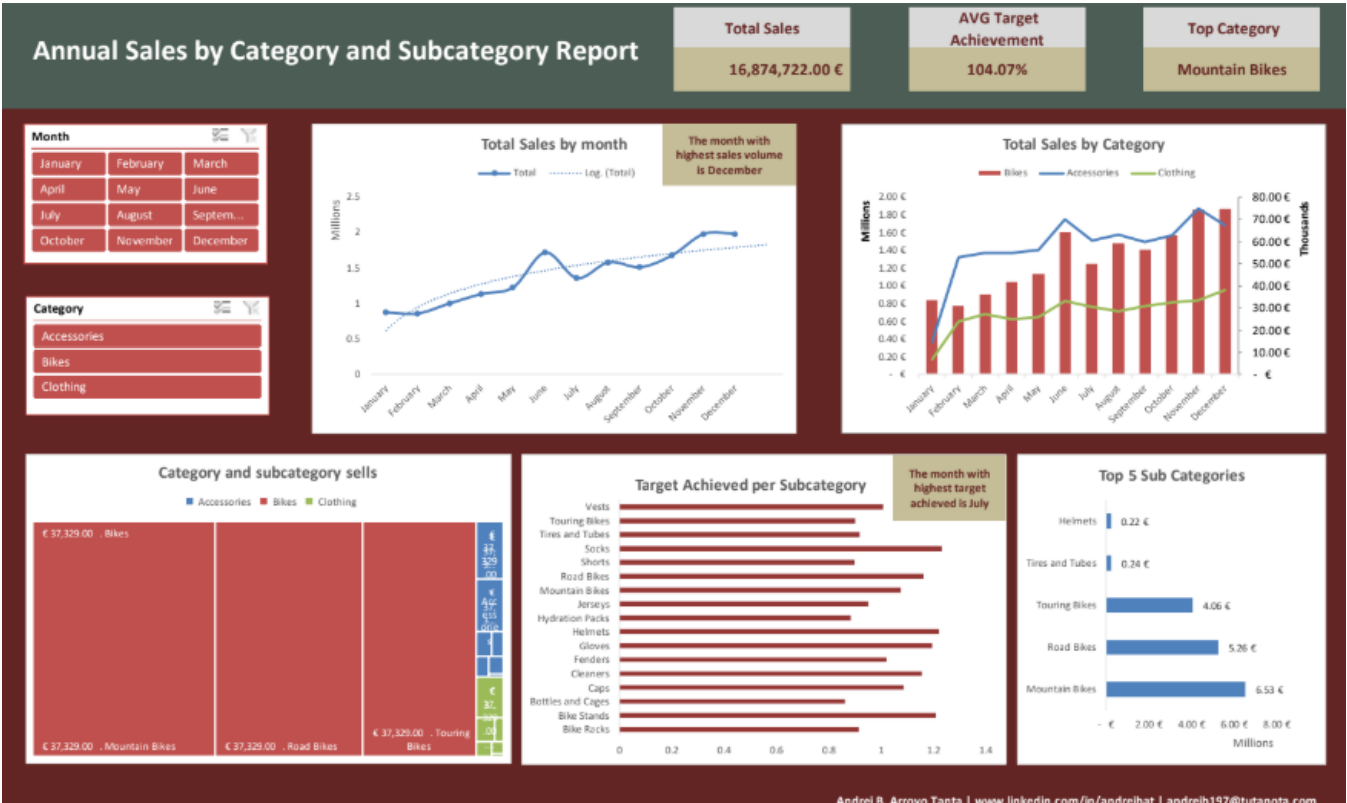
- ▶ **Portfolio optimization:** instantly identifies which brands (e.g., Bacci, Lala) lead revenue, enabling inventory and marketing decisions based on facts, not intuition.
- ▶ **Talent management (performance tracking):** accurately quantifies volume processed per agent (e.g., 411 items handled by top performers), supporting fair incentive schemes and bottleneck detection.
- ▶ **Operational scalability:** folder-based architecture auto-updates the model as new files arrive, cutting report preparation time by **90%**.

3. Technology Stack: Data Engineering with Power Query

- ▶ **Power Query (M language):** core engine for normalizing agent names, Fill Down brand hierarchies, and cleaning inconsistent barcodes.
- ▶ **Multi-source transformation:** complex joins between TXT files and Excel tables to enrich transactions with store and product metadata.
- ▶ **Dynamic data modeling:** custom revenue calculation columns and logical filters to remove duplicate or corrupt records (data cleansing).
- ▶ **Business Intelligence (PivotTables):** dynamic reports segmenting business reality by brand, agent, and week.

Sales Performance Optimizer: Data Architecture for Goal Control and Consumption Trend Analysis

Built a data transformation engine (ETL) that automates the normalization of complex sales records, identifying performance gaps against targets and product seasonality with full operational accuracy.



Annual Sales by Category and Subcategory Report — €16.87M total sales, 104.07% average target achievement

1. Business Need

Retail and distribution constantly face "trapped data": financial reports designed to be read, but not analyzed. The critical problem was the inflexibility of source data — empty rows, irregular headers, and months arranged as columns — which blocked dynamic analysis. Without a normalizing architecture, companies suffer "analytics paralysis," unable to calculate vital metrics like quota compliance or historical trends.

2. Strategic Importance & Impact

- ▶ **Business effectiveness monitoring (% target achieved):** instantly flags which subcategories exceed targets and which need intervention, optimizing resource allocation.
- ▶ **Seasonality & time flow analysis:** converts month names into chronological dimensions, enabling stacked area charts that reveal demand peaks in categories like Accessories and Clothing.
- ▶ **Data scalability & automation:** a cleansing pipeline auto-integrates new data, removing manual spreadsheet risk and ensuring monthly report integrity.

3. Technology Stack & Technical Competencies

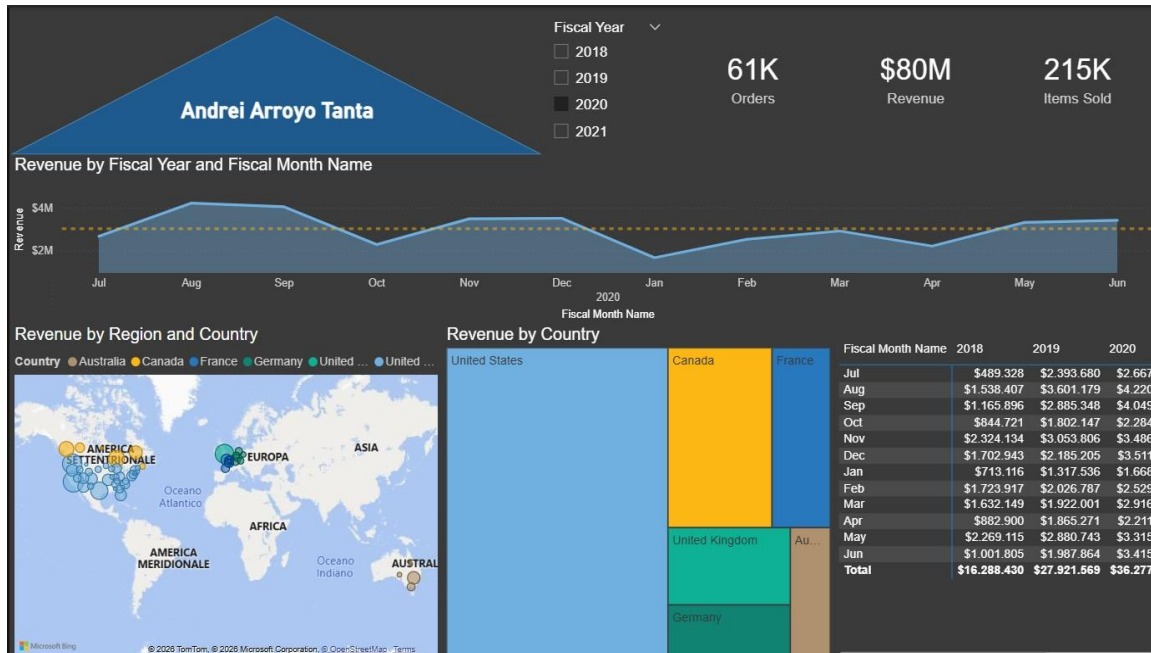
- ▶ **Data engineering with M language (Power Query):** deep-cleansing scripts, Fill Down logic for categories, removal of structural noise.

- ▶ **Normalization & unpivoting:** converted horizontal matrices into a tabular Fact Table model, enabling PivotTable calculation engines.
- ▶ **Temporal dimension transformation:** M-language logic converting text month labels into numeric values for precise chronological ordering.
- ▶ **Business KPI calculation:** custom columns for goal-fulfillment analysis (Sales / Monthly Target), formatted for direct executive interpretation.
- ▶ **Dynamic trend visualization:** stacked-area Pivot Charts with advanced category segmentation, highlighting critical periods like February.

PROJECT 4

Dynamic Fiscal Analytics: End-to-End Architecture for Revenue Control and Operational Optimization with Power BI

Designed an advanced BI infrastructure automating data ingestion and critical KPI calculation, ensuring accurate financial reporting aligned with the corporate fiscal calendar and geographic reality of the business.



Dynamic Fiscal Analytics Dashboard — 61K orders, \$80M revenue, 215K items sold across fiscal years 2018-2021

1. Business Need

Most companies struggle with "fiscal myopia" — the inability to align transactional data with real financial cycles, which rarely match the calendar year. This project solves the chaos of manually cleaning data scattered across customers, orders, and staff, where unresolved geographic ambiguities lead management to make decisions on fragmented data and misallocate budgets and sales goals.

2. Strategic Importance & Impact

- ▶ **Comparative temporal intelligence:** Time Intelligence functions surface Month-to-Month (MoM) growth and percentage variation with precision; Waterfall Charts show exactly how revenue accelerated or slowed.
- ▶ **Hierarchical & drill-down navigation:** fiscal and geographic hierarchies let leaders move from global revenue to a specific SKU or county in seconds.
- ▶ **Data-driven culture with professional UX:** automatic conditional formatting reduces cognitive load, letting non-technical users instantly spot risk trends (red alerts) and opportunities (green arrows).

3. Technology Stack & Technical Competencies

- ▶ **Advanced ETL (Power Query):** raw data cleansing, geo-disambiguation via logical concatenation, and dynamic personnel-dimension filtering.
- ▶ **Multidimensional modeling (star schema, Power Pivot & DAX):** 1:N relational model with a Dynamic Calendar Table built in DAX, configured to fiscal cycles and dynamic dates (Max Date + 60 days).
- ▶ **DAX engineering:** complex temporal-intelligence calculations (PREVIOUSMONTH, SAMEPERIODLASTYEAR) and DIVIDE with error handling for revenue segmentation.

- **Visualization & UI/UX:** synchronized Slicers, corporate hex color themes, and third-party visuals for advanced distribution analytics.

Technical Skills Demonstrated

Power Query (M)	ETL automation, Fill Down logic, multi-source joins, data cleansing, unpivoting
Power Pivot & DAX	Star schema modeling, CALCULATE, time intelligence (SAMEPERIODLASTYEAR, PREVIOUSMONTH), DIVIDE error handling
Data visualization	Interactive dashboards, Sunburst & Waterfall charts, synchronized Slicers, conditional formatting
Business analytics	KPI design, target-achievement tracking, YoY growth analysis, fiscal calendar modeling

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